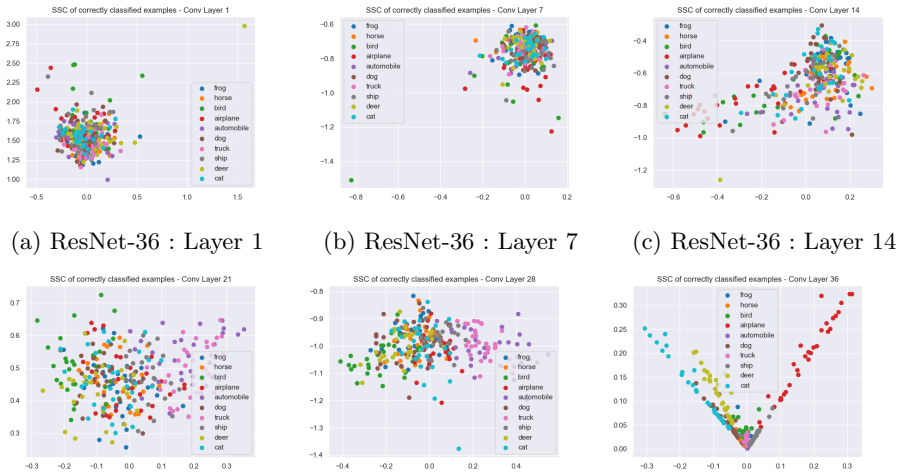


(a) ResNet-36 : Layer 1 (b) ResNet-36 : Layer 7 (c) ResNet-36 : Layer 14

(d) ResNet-36 : Layer 21 (e) ResNet-36 : Layer 28 (f) ResNet-36 : Layer 36

Fig. 8: Left to Right : Spectral Embeddings of Layer-wise Affinity graphs learned via a ResNet-36 - Data Set : CIFAR 10

In Figure 9a we show the input image of a cat that was incorrectly classified as a bird with a softmax score of 0.97. Figure 9b shows the normalised class-wise distribution of affinity scores for this input summed over all the images. This plot clearly shows the three strongest classes for which the given input has a strong affinity for, namely, bird, airplane and cat, in that order. At this juncture, we make a note that the predicted label, though incorrectly, of this input by the network seems to be the class for which this input has the highest affinity for as determined by Sparse Subspace Clustering Algorithm. Figure 9c- Figure 9j show the top-8 images, in descending order, for which the given input of cat has the highest affinity for.

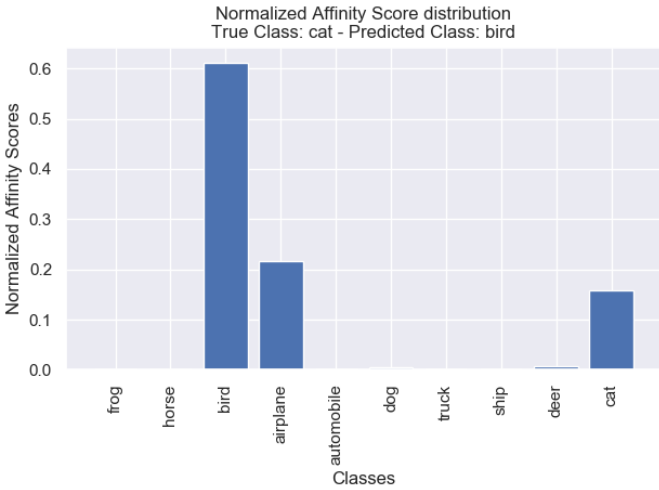
Continuing discussion on the previous observation, we further expand on that observation in Table 1, where we present the accuracy of the network on the testing set in two scenarios. The first scenario assigns an output label to a given input based on the highest aggregate class-wise affinity score for that input and the second scenario assigns the same output label to the input as the networks prediction through it's classification layers. The results of those are presented in columns 2 and 3 labeled 'SSC Label' and 'Network Prediction' respectively, in Table 1. We observe that both these 'accuracies' turn out to be 95.8%, Please be advised that the equality of values is a coincidence. Thus, given the activations of the final convolutional layer for the entire testing set we observe that the SSC based Affinity scoring assigns labels which are competitive with what the classification layers of the network could learn. This is further demonstrated in column 4 of Table 1 which shows a 98.3% agreement between one to one comparisons of network assigned labels with SSC derived labels. Row

3 and row 4 show the same metrics for other 2 networks, on CIFAR-10. Such a high annotation agreement between SSC and network prediction indicates a tendency of neural networks to separate out data points belonging to different classes into disjoint sections of the latent embedding space.

Table 1: **SSC Labels vs. Network Labels**: Comparison of Accuracy and Correlation on the test set

Accuracy of Labels when compared to ground truth (CIFAR-10)			
Network	SSC Label	Network Prediction	Correlation
ResNet-34	95.8%	95.8%	98.3%
ResNet-18	95.4%	96.1%	97%
ResNet-50	90.6%	94.8%	93.5%

(a) Input



(b) Neighbourhood Distribution (Normalised)

(c) Airplane

(d) Cat

(e) Bird

(f) Bird

(g) Airplane

(h) Bird

(i) Bird

(j) Bird

Fig. 9: 9(a): A Cat, classified as a bird with a softmax score of 0.97. 9(b): Distribution of it's Neighbourhood Affinity Scores (Normalised). 9(c): Airplane. 9(d): Cat. 9(e): Bird. 9(f): Bird. 9(g): Airplane. 9(h): Bird. 9(i): Bird. 9(j): Bird. Network: ResNet-36 - Data Set : CIFAR 10

7 Conclusions

Our work proposes a two-step framework to analyze deep neural networks. This framework combines Sparse Subspace Clustering and Centered Kernel Alignment to provide the ability to analyze the network on a macro and micro level. The macro analysis helps in visualising network architectures as a function of network depth, width, training epochs and training data quantity and provides an insight into network architecture and training dynamics of the network. It also provides a framework to compare two architecturally different neural network based on a common set of inputs. The framework also provides the ability to micro analyze the network in the form of instance based interpretability by providing a measure of a degree of closeness each input has to decision boundaries for different classes in the loss landscape of a network.

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